Year 8 Science – Booklet 23: Biology

Inheritance (Inheritance and Genetics) — ANSWERS

> Note: this booklet's content is identical to Year 8 Booklet 16 (and Year 9 Booklet 8, Genetics & DNA), so the same worked answers apply throughout.

Quick Test (p.3)

1. In the nucleus of the cell, carried on the chromosomes.
2. 46 (23 pairs).
3. 23.
4. 20.
5. bb (black hair is recessive, so a person showing it must have two copies of the recessive allele).

Section A – Multiple Choice (5 marks)

1. a) on a chromosome
2. d) 2
3. c) radiation
4. c) a disease of the red blood cells
5. b) 1

Section B (13 marks)

1. Fill in the gaps (8 marks)

Inside a cell is the nucleus, which controls all our inherited features. Inside the nucleus are thread-like **chromosomes**. On the chromosomes are **pairs** of **genes**. Humans have 46 chromosomes. The genes are made up of a chemical called **DNA**. We inherit two genes for every feature. Genes can be more powerful and called **dominant** or less powerful and called **recessive**. Occasionally during the inheritance process things can go wrong. This is called a **mutation**. The chances of this happening are increased by exposure to certain chemicals and **radiation**.

2. Red rose × white rose → all red offspring

a). Red (since crossing red with white gave only red offspring, red must be dominant).

b). Rr (heterozygous – one dominant R from the red parent, one recessive r from the white parent).

c) Punnet square for Rr × Rr:

	R	r
R	RR	Rr
r	Rr	rr

Offspring: RR, Rr, Rr, rr → 3 red (RR, Rr, Rr) : 1 white (rr), matching the observed “three times as many red roses as white roses”.

Section C (7 marks)

1. Genes and appearance

a). The environment (e.g. diet, lifestyle) also affects what we look like, not just our genes.

b). The nucleus.

c). 46 (23 pairs).

d). 20.

2. Punnet square (mother bb × father Bb): offspring Bb, bb, Bb, bb

a). 2 out of 4 (50%) brown-eyed children.

b). 2 out of 4 (50%) blue-eyed children.

c). Bb (heterozygous – brown eyes, but carrying one recessive blue allele).